

Problem A: Binary search

Input file: **standard input** | Output file: **standard output** | Time limit: **1 sec** | Memory: **256 MB**

You are given a sorted array and a number x . Determine whether x occurs in the array.

This is the warm-up for the technique: the array here is small, but write the search so that it halves the range at every step rather than scanning the elements one by one — the same code will be needed later on arrays where scanning is far too slow.

Input Format

The first line contains one integer n — the size of the array ($1 \leq n \leq 100$).

The second line contains n integers $a_1 \leq a_2 \leq \dots \leq a_n$ — the elements of the array in non-decreasing order ($1 \leq a_i \leq 20000$). Equal elements are allowed.

The third line contains one integer x — the number to look for ($-20000 \leq x \leq 20000$).

Output Format

Print Yes if x occurs in the array, and No otherwise.

Examples

Input

```
5
1 2 3 4 5
1
```

Output

```
Yes
```

Input

```
5
1 2 3 4 5
7
```

Output

```
No
```

Problem B: Patchwork Staccato I

Input file: **standard input** | Output file: **standard output** | Time limit: **4 sec** | Memory: **256 MB**

You are given array $\mathbf{a}$ ($1 \leq a_i \leq 10^9$) of length n ($1 \leq n \leq 100$) and q ($1 \leq q \leq 100$) queries. In query i you are given two pairs of segments l_1, r_1, l_2, r_2 ($1 \leq l_1 \leq r_1 \leq 10^9, 1 \leq l_2 \leq r_2 \leq 10^9$), find number of indices c ($1 \leq c \leq n$) for which one of the following conditions is satisfied: $l_1 \leq a_c \leq r_1$ or $l_2 \leq a_c \leq r_2$.

Input Format

The first line contains two integers n and q ($1 \leq n \leq 100, 1 \leq q \leq 100$).

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$) – the array itself, given in arbitrary order.

Each of the next q lines contains 4 integers l_1, r_1, l_2, r_2 – one query.

Output Format

Output q lines the answer to each query in the order they are given.

Examples

Input

```
7 3
21 1 2 3 5 8 13
1 5 13 21
1 1 2 3
1 3 2 8
```

Output

```
6
3
5
```

Problem C: Debugging

Input file: **standard input** | Output file: **standard output** | Time limit: **1 sec** | Memory: **256 MB**

Jonathan almost finished his project by Object-Oriented Programming course. His code consists of N consecutive blocks, each of them consists of a certain amount of lines. Unfortunately, Jonathan made a lot of mistakes. Compiler showed that Jonathan made M mistakes, each of them is described by the number of line where this mistake was made. To debug his project faster, Jonathan wants to define number of block in which he made a mistake. Please, help Jonathan debug his project before deadline will expire.

Input Format

First line consists of integers N and M — number of blocks and mistakes ($1 \leq N, M \leq 2 \cdot 10^5$).

The second line contains N integers a_i — number of lines in the i -th block ($1 \leq a_i \leq 10^4$).

Each of the next M lines contains one integer b_i — number of line where the i -th mistake was made ($1 \leq b_i \leq a_1 + a_2 + \dots + a_N$). In other words, every mistake is guaranteed to fall inside the code, so the answer always exists.

Output Format

Print M lines, the i -th line must contain the number of block in which the i -th mistake was made.

Examples

Input

```
2 1
3 4
5
```

Output

```
2
```

Input

```
3 3
5 7 6
5
10
15
```

Output

```
1
2
3
```

Problem D: Win me if you can!

Input file: **standard input** | Output file: **standard output** | Time limit: **1 sec** | Memory: **256 MB**

Mark is going to fight for Fight Club. There were N competitors with powers. There will be P rounds to fight and in each round Mark's power will be changed. With power M , Mark can kill all the competitors whose power is equal to or less than his. Round by round, all the competitors who are dead in the previous round will be reborn. Such that in each round there will be N competitors to fight. As Mark is tired, please, help him to count the number of competitors that he can win in each round and the total sum of their powers.

Input Format

The first line contains an integer N ($1 \leq N \leq 2 \cdot 10^5$) – the number of competitors without Mark.

Next line contains N integers a_i ($1 \leq a_i \leq 10^3$) – powers of these competitors.

The third line contains one integer P ($1 \leq P \leq 2 \cdot 10^5$) – the number of rounds.

Each of the next P lines contains an integer p_i ($1 \leq p_i \leq 10^3$) – power of Mark at each round.

Output Format

On each of the P lines print two integers: how many competitors Mark beats in this round and the total sum of their powers.

Examples

Input

```
7
7 9 1 8 2 6 2
2
4
8
```

Output

```
3 5
6 26
```

Problem E: Patchwork Staccato II

Input file: **standard input** | Output file: **standard output** | Time limit: **6 sec** | Memory: **256 MB**

You are given array $\mathbf{a}$ ($1 \leq a_i \leq 10^9$) of length n ($1 \leq n \leq 10^5$) and q queries ($1 \leq q \leq 10^5$). In query i you are given two pairs of segments l_1, r_1, l_2, r_2 ($1 \leq l_1 \leq r_1 \leq 10^9, 1 \leq l_2 \leq r_2 \leq 10^9$), find number of indices c ($1 \leq c \leq n$) for which one of the following conditions is satisfied: $l_1 \leq a_c \leq r_1$ or $l_2 \leq a_c \leq r_2$.

Input Format

The first line contains two integers n and q ($1 \leq n \leq 10^5, 1 \leq q \leq 10^5$).

The second line contains n integers $a_1, a_2, \dots, a_n$ — the array itself, given in arbitrary order.

Each of the next q lines contains 4 integers l_1, r_1, l_2, r_2 — one query.

Output Format

Output q lines the answer to each query in the order they are given.

Examples

Input

```
7 3
21 1 2 3 5 8 13
1 5 13 21
1 1 2 3
1 3 2 8
```

Output

```
6
3
5
```

Notes: The array $\mathbf{a}$ is given in arbitrary order. Sort it first, after that the answer to a query is found by binary search over the sorted array. Both segments are closed: the elements equal to r_1 or r_2 are counted too, and an element that falls into both segments is counted only once.

Problem F: Robin Hood stealing the Gold

Input file: **standard input** | Output file: **standard output** | Time limit: **1 sec** | Memory: **256 MB**

Robin Hood wants to steal the golden bars from the bank of High Sheriff aiming to distribute them to poor local people. There are N bags of golden bars, the i -th bag has $bags[i]$ bars. Sheriff has gone and will return in H hours.

Robin can steal K bars per hour. Each hour, he chooses a single bag of golden bars, and steals K bars from that bag. If there are less than K bars in the bag, he steals them all, and won't steal any more during this hour.

Robin Hood wants to steal all of the golden bars before the Sheriff comes back. Return the minimum number K such that Robin can steal ALL of the golden bars within H hours.

Input Format

The first line of the input contains two space-separated integers N ($1 \leq N \leq 10^4$), H ($N \leq H \leq 10^9$), the number of bags of golden bars and the number of hours for which Sheriff has gone. The next line contains N space-separated integers ($1 \leq bags[i] \leq 10^9$) denoting the number of golden bars in each bag.

Output Format

Print the minimum number K such that Robin Hood can steal all of the N golden bars within the limit of H hours.

Examples

Input

```
4 8
3 6 7 11
```

Output

```
4
```

Input

```
5 5
30 11 23 4 20
```

Output

```
30
```

Input

```
5 6
30 11 23 4 20
```

Output

```
23
```

Problem G: Cutting the Ropes

Input file: **standard input** | Output file: **standard output** | Time limit: **1 sec** | Memory: **256 MB**

Aisha is preparing decorations for the university open day. She has n ropes; the i -th of them is a_i centimetres long.

For the decoration she needs exactly k pieces of rope, and all of them must be of the **same** length. A piece is always cut out of a single rope — ropes may not be tied together. Whatever is left over after cutting is simply thrown away, and it is fine to leave a whole rope untouched.

Aisha wants the pieces to be as long as possible. Find that length.

Note that the answer does not have to be an integer number of centimetres: the rope may be cut at any point.

Input Format

The first line contains two integers n and k — the number of ropes and the number of pieces Aisha needs ($1 \leq n \leq 10^5, 1 \leq k \leq 10^5$).

The second line contains n integers $a_1, a_2, \dots, a_n$ — the lengths of the ropes in centimetres ($1 \leq a_i \leq 10^9$).

Output Format

Print one real number — the greatest length such that k pieces of that length can be cut out of the ropes.

Your answer is accepted if its absolute or relative error does not exceed 10^{-6} .

Print the answer with at least 9 digits after the decimal point (`printf("%.9f", ans)` in C++, `print(f"{ans:.9f}")` in Python): the default output of `cout << ans` keeps only 6 significant digits and is not enough.

Examples

Input

```
4 11
802 743 457 539
```

Output

```
200.5000000000
```

Input

```
3 3
5 5 5
```

Output

```
5.0000000000
```

Input

```
3 4
```

Problem H: K-subarray

Input file: **standard input** | Output file: **standard output** | Time limit: **1 sec** | Memory: **256 MB**

You are given an array of non negative integers and a number **k**. Let's define subarray as a non-empty *consecutive* elements of an array. Among all subarrays of the given array, find the one, such that sum of its elements is no less than **k** and such that this subarray would contain minimum possible number of elements.

Input Format

First line contains two space separated numbers **n k** – number of elements in given array and number that was mentioned above, respectively ($1 \leq n \leq 10^5$, $0 \leq k \leq 10^9$).

Second line contains **n** space separated numbers **a₁, a₂, ..., a_n** – given array ($0 \leq a_i \leq 10^4$).

It is guaranteed that at least one subarray's sum is not less than **k**.

Output Format

Output a single integer **x** – the number of elements in the shortest subarray whose sum is not less than **k**.

Examples

Input

```
3 12
3 5 7
```

Output

```
2
```

Input

```
6 19
3 6 1 4 5 2
```

Output

```
5
```

Notes In the first test case we have three elements. Subarrays are [3], [3, 5], [3, 5, 7], [5], [5, 7], [7]. Only two subarrays have sum that is not less than **k = 12**: [3, 5, 7], [5, 7]. Out of these two subarrays, [5, 7] has minimum possible length of 2.

Hint For a fixed left end of subarray, sums of subarray increase, if we increase number of elements in it. So we can do binary search on right end of subarray for a fixed left end.

In order to quickly know sum of subarray we can calculate prefix sums. For example for the given array from the first test case prefix sums would be [3, 8, 15]. In order to get sum of second and third element, from prefix sum at position 3 we subtract prefix sum at position 1: $15 - 3 = 12$.

Scoring Group 1 (40 points) and group 2 (52 points): random and adversarial tests.
Group 3 (8 points): corner cases (**k** = 0, answer equal to 1, answer equal to **n**).

Problem I: Oshiete oshiete yo sono shikumi wo

Input file: **standard input** | Output file: **standard output** | Time limit: **1 sec** | Memory: **256 MB**

There is only one road and n houses in Tokyo, and all the houses are on this road. Houses are numbered from 1 to n and appear in this order. There are a_i ghouls living in the i -th house. Due to the RC-cells infection, $k-1$ roadblocks need to be installed between houses in Tokyo, so that k blocks of houses are detached. Kaneki Ken wants to divide ghouls so that the maximum number of ghouls over blocks (consecutive houses detached by roadblocks) is minimal. Help Kaneki find this number.

Input Format

The first line contains integers n and k ($1 \leq k \leq n \leq 10^5$). The second line contains the elements of the array a_i ($1 \leq a_i \leq 10^9$).

Output Format

Print one number the minimum possible maximum number of ghouls on a section of the roadblock.

Examples

Input

```
10 3
3 4 2 1 3 4 5 2 2 3
```

Output

```
10
```

Input

```
10 4
3 1 2 4 10 8 4 2 5 3
```

Output

```
12
```

Problem J: Jonathan the Farmer

Input file: **standard input** | Output file: **standard output** | Time limit: **5 sec** | Memory: **256 MB**

Jonathan is the Farmer whose household was damaged by a huge hurricane. He lost majority of his cattle. One day he walked near his farm and observed that there are N sheep on the field. Each sheep is always grazing inside some rectangular area. Jonathan remembered such areas for each sheep. When he came home, he decided to build a paddock to catch at least K sheep (to catch a sheep Jonathan must cover sheep's pasture fully). Jonathan prefers squares rather than usual rectangles, therefore he wants to build square paddock with the corner at point $(0,0)$. Material for paddock costs money, so Jonathan wants to minimize the length of paddock side. Help him find this length.

Input Format

The first line of the input contains two integers N and K ($1 \leq K \leq N \leq 2 \cdot 10^5$) – number of sheep grazing in the field and the number of sheep Jonathan wants to catch.

Each of the next N lines contains four integers $x_{i,1}, y_{i,1}, x_{i,2}, y_{i,2}$ ($1 \leq x_{i,1} < x_{i,2} \leq 10^9, 1 \leq y_{i,1} < y_{i,2} \leq 10^9$) – coordinates of bottom-left and top-right corners of the i -th sheep's pasture.

Output Format

Find the minimum length of square paddock such that at least K sheep's pastures fit there.

Examples

Input

```
10 7
5 1 7 8
1 3 5 4
5 8 8 10
7 1 8 5
9 5 1 10
4 4 7 5
1 6 7 7
9 7 5 10
4 8 5 9
4 2 5 3
```

Output

```
9
```

Problem K: Snake

Input file: **standard input** | Output file: **standard output** | Time limit: **1 sec** | Memory: **256 MB**

Write a program that, for several given values, prints their coordinates in a *snake* array of size $n \times m$.

A snake array is filled like this:

- every column strictly decreases from top to bottom: $a_{i,j} > a_{i+1,j}$
- every **even** row strictly decreases from left to right: $a_{i,j} > a_{i,k}$ for $j < k$;
- every **odd** row strictly increases from left to right: $a_{i,j} < a_{i,k}$ for $j < k$.

Rows are numbered from 0 to $n - 1$ from top to bottom, columns from 0 to $m - 1$ from left to right.

Here is an example of 3×4 Snake array

```
25 23 20 19
13 15 17 18
12 10 9 8
```

Input Format

The first line of input contains a single integer t — the number of values whose coordinates you must find ($1 \leq t \leq 10^4$).

The next line contains t integers — the values themselves ($-10^7 \leq \text{each value} \leq 10^7$).

The next line contains two integers n and m — the number of rows and the number of columns ($1 \leq n, m \leq 800$).

Each of the next n lines contains m integers — the snake array itself ($-10^7 \leq a_{i,j} \leq 10^7$, $0 \leq i < n$, $0 \leq j < m$). It is guaranteed that the array really is a snake array.

Output Format

Print t lines, one per requested value, in the order the values are given.

For a value that occurs in the array print two integers — the number of its row and the number of its column. **Rows and columns are numbered from zero.** If the value does not occur in the array, print -1 instead.

Examples

Input

```
5
10 15 13 8 23
3 4
25 23 20 19
13 15 17 18
12 10 9 8
```

Output

```
2 1
1 1
```